polybench/seidel-2d · seidel-2d.c · full · claude-haiku-4-5-20251001 · no-change · RIGHT = a rewrite the gate rejected at 'correctness'

−3 / +30 lines of 115 · 4 parallel region(s) added · 0 of 12 candidate(s) kept · RESTRUCTURING (19 code line(s) added, 3 removed) · restructuring level 0

BEFORE

```
58                DATA_TYPE POLYBENCH_2D(A,N,N,n,n))
59  {
60    int t, i, j;
61
62  #pragma scop
63    for (t = 0; t <= _PB_TSTEPS - 1; t++)
64      for (i = 1; i<= _PB_N - 2; i++)
65        for (j = 1; j <= _PB_N - 2; j++)
66      A[i][j] = (A[i-1][j-1] + A[i-1][j] + A[i-1][j+1]
67             + A[i][j-1] + A[i][j] + A[i][j+1]
68             + A[i+1][j-1] + A[i+1][j] + A[i+1][j+1])/9.0;
69  #pragma endscop
70
71  }
72
73
```

A REWRITE THE GATE REJECTED AT 'CORRECTNESS'

```
58                DATA_TYPE POLYBENCH_2D(A,N,N,n,n))
59  {
60    int t, i, j;
61
62  #pragma scop
63    for (t = 0; t <= _PB_TSTEPS - 1; t++) {
64      // Pass 0: cells where both i and j are odd
65      #pragma omp parallel for collapse(2)
66      for (i = 1; i <= _PB_N - 2; i += 2)
67        for (j = 1; j <= _PB_N - 2; j += 2)
68      A[i][j] = (A[i-1][j-1] + A[i-1][j] + A[i-1][j+1]
69             + A[i][j-1] + A[i][j] + A[i][j+1]
70             + A[i+1][j-1] + A[i+1][j] + A[i+1][j+1])/9.0;
71
72      // Pass 1: cells where i is odd and j is even
73      #pragma omp parallel for collapse(2)
74      for (i = 1; i <= _PB_N - 2; i += 2)
75        for (j = 2; j <= _PB_N - 2; j += 2)
76      A[i][j] = (A[i-1][j-1] + A[i-1][j] + A[i-1][j+1]
77             + A[i][j-1] + A[i][j] + A[i][j+1]
78             + A[i+1][j-1] + A[i+1][j] + A[i+1][j+1])/9.0;
79
80      // Pass 2: cells where i is even and j is odd
81      #pragma omp parallel for collapse(2)
82      for (i = 2; i <= _PB_N - 2; i += 2)
83        for (j = 1; j <= _PB_N - 2; j += 2)
84      A[i][j] = (A[i-1][j-1] + A[i-1][j] + A[i-1][j+1]
85             + A[i][j-1] + A[i][j] + A[i][j+1]
86             + A[i+1][j-1] + A[i+1][j] + A[i+1][j+1])/9.0;
87
88      // Pass 3: cells where both i and j are even
89      #pragma omp parallel for collapse(2)
90      for (i = 2; i <= _PB_N - 2; i += 2)
91        for (j = 2; j <= _PB_N - 2; j += 2)
92      A[i][j] = (A[i-1][j-1] + A[i-1][j] + A[i-1][j+1]
93             + A[i][j-1] + A[i][j] + A[i][j+1]
94             + A[i+1][j-1] + A[i+1][j] + A[i+1][j+1])/9.0;
95    }
96  #pragma endscop
97
98  }
99
100
```

file

red = removed · green = added · yellow = parallel region (#pragma omp) · right strip = the whole file: green where changed, yellow where a parallel region now is